

Retrieval augmented generation, or RAG, is a pattern for building assistants that answer questions from supplied documents. A RAG system first loads a document, extracts readable text, and splits that text into chunks. Each chunk is converted into a searchable representation called an embedding.

When a user asks a question, the app embeds the question and compares it with the document chunks. The most relevant chunks are retrieved and passed to the answering layer. This keeps answers grounded in the uploaded document instead of relying only on a model's memory.

This project uses Streamlit for the user interface. The user uploads a PDF, TXT, or Markdown file, then asks a question. If the question box is left blank, the app summarizes the document. For retrieval, the project uses TF-IDF vectors and cosine similarity, which are lightweight and easy to deploy.

Chunking is important because very long documents are difficult to search and too large to send to a language model at once. The app uses overlapping word chunks so important context is less likely to be cut off at chunk boundaries.

For public deployment, the app can use a hosted OpenAI-compatible model through Groq. For local demos, it can also call Ollama on the same machine. If no model provider is configured, the app still returns an extractive answer using the retrieved chunks.